Yunyi Cheng

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SUMMARY

Algorithm and machine learning engineer building sensing algorithms, predictive models, and production data systems for wearable and clinical time-series data. Experienced in signal processing, feature engineering, model development and evaluation, and backend implementation using Python, PyTorch, SQL, and R. Translates open-ended scientific problems into reproducible ML pipelines, rigorously validated algorithms, and reliable user-facing health and performance features.

EDUCATION

Harvard Medical School May 2026
Master of Medical Sciences in Biomedical Informatics GPA: 3.75/4.00

University of Toronto June 2024
Honours Bachelor of Science in Computer Science and Bioinformatics & Computational Biology GPA: 3.73/4.00

EXPERIENCE

Algorithm Engineer - Wearable Sensing | STRIV Aug 2026 – Present

- Develop signal-processing, statistical, and machine-learning methods for multivariate pressure and motion time series, transforming raw wearable signals into user-facing running metrics and insights.
- Define and validate biomechanics metrics using real-world runs and reference data, evaluating reliability across runners, running conditions, and levels of sensor data quality.
- Own the algorithm lifecycle from data ingestion and exploratory analysis through prototyping, offline evaluation, backend implementation, and production monitoring.
- Collaborate across biomechanics, product, mobile, and backend engineering to translate research concepts into reliable production features.

Machine Learning Researcher | Beth Israel Deaconess Medical Center Feb 2025 – Jul 2026

- Developed multimodal risk-stratification models integrating EEG time-series features, imaging findings, clinical variables, and unstructured notes, improving enrichment for anti-epileptogenic drug trial enrollment by approximately 50% compared with random selection.
- Built reproducible R and Python workflows for survival analysis, competing-risks modeling, stratified train-test evaluation, and cross-validation across baseline, imaging, EEG, and combined model tiers.
- Applied NLP methods to electronic medical records and clinical notes to extract outcomes and structured variables for cohort construction and downstream predictive modeling.
- Evaluated model performance across modalities and patient subgroups using closed-group testing, false-discovery-rate control, benchmarking, and quantitative visualization.

Computational Biology Research Assistant | Röst Lab, University of Toronto May 2023 – June 2024

- Developed a Graph Transformer-based model to generate synthetic mass spectra from molecular structures, supporting data augmentation for proteomics research.
- Engineered molecular graph representations and implemented multi-head attention and multilayer perceptron modules for high-resolution spectral-intensity prediction.
- Validated predictions against experimental datasets to assess model reliability, robustness, and generalization.

Software Engineer | Intel of Canada, Ltd. Jul 2022 – Sep 2023

- Developed SQL data logic, reporting tables, and Power BI dashboards for cross-functional project tracking and operational decision-making.
- Debugged dashboard performance and evaluated gRPC integration for API-based software-testing and workflow-automation systems.

Genetics Researcher | ANNB Lab, University of Toronto May 2022 – May 2023

- Designed a computational optimization pipeline for tile placement and overhang selection in deep mutational scanning experiments, increasing on-target reactions while reducing experimental errors.
- Implemented greedy optimization and Metropolis-Hastings sampling in Python and developed an R package for reproducible assembly design; contributed to a bioRxiv preprint.

SELECTED AI/ML PROJECTS

LungLDM: Conditional Latent Diffusion for Lung Cancer Histopathology, BMI702 2025

- Built a conditional latent diffusion model to generate lung cancer histopathology images conditioned on GPT-4o-summarized pathology reports, tumor scores, and tumor-infiltrating lymphocyte scores.
- Integrated pathology text, image-derived labels, and clinical features into a multimodal generative modeling framework for high-resolution histopathology image synthesis.
- Evaluated generated images using structural similarity metrics, achieving an SSIM score of approximately 0.915.

Pisces: Graph-Based Cell–Cell Communication Modeling, MIT 6.8701 2024

- Built a scalable graph-based machine learning pipeline using Graph Attention Networks to model cell–cell communication from MERFISH spatial transcriptomics data.
- Integrated MERFISH data with heterogeneous biological networks, including gene regulatory networks, protein–protein interaction networks, and ligand–receptor relationships.
- Designed graph representations to connect spatial gene expression, cell identity, and prior biological knowledge for downstream biological interpretation.

SKILLS

Programming & ML Tools: Python, PyTorch, Scikit-learn, NumPy, Pandas, R, SQL, C

Wearable Sensing & Algorithms: Signal Processing, Time-Series Analysis, Multivariate Sensor Data, Wearable Algorithms, Biomechanics, Algorithm Development, Algorithm Validation

Machine Learning: Predictive Modeling, Deep Learning, Multimodal Learning, Graph Neural Networks, Transformers, Diffusion Models, Natural Language Processing

Statistics & Evaluation: Statistical Modeling, Survival Analysis, Competing Risks, Cross-Validation, Hypothesis Testing, False Discovery Rate Control, Benchmark Design

Data & ML Engineering: Reproducible ML Pipelines, Feature Engineering, Data Processing, ETL, Relational Databases, API Integration, Git, Linux/Unix, HPC/GPU Computing